

EECS 836 HW3

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Problem 1: Query-Key-Value Mechanics in Self-Attention

Solution 1

$$\alpha_{2,1} = \frac{\mathbf{q}_2^\top \mathbf{k}_1}{\sqrt{d}}, \quad \alpha_{2,2} = \frac{\mathbf{q}_2^\top \mathbf{k}_2}{\sqrt{d}}, \quad \alpha_{2,3} = \frac{\mathbf{q}_2^\top \mathbf{k}_3}{\sqrt{d}}$$

$$\mathbf{q}_2 = W_q \mathbf{a}_2 = \begin{bmatrix} 1 & 0 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} (1 \cdot 4) + (0 \cdot 3) \\ (2 \cdot 4) + (9 \cdot 3) \end{bmatrix} = \begin{bmatrix} 4 \\ 35 \end{bmatrix}$$

$$\mathbf{k}_1 = W_k \mathbf{a}_1 = \begin{bmatrix} 0 & 1 \\ 6 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} (0 \cdot 5) + (1 \cdot 1) \\ (6 \cdot 5) + (0 \cdot 1) \end{bmatrix} = \begin{bmatrix} 1 \\ 30 \end{bmatrix}$$

$$\alpha_{2,1} = \frac{\mathbf{q}_2^\top \mathbf{k}_1}{\sqrt{d}} = \begin{bmatrix} 4 & 35 \end{bmatrix} \begin{bmatrix} 1 \\ 30 \end{bmatrix} = (4 \cdot 1) + (35 \cdot 30) = 4 + 1050 = \frac{1054}{\sqrt{2}}$$

$$\mathbf{k}_2 = W_k \mathbf{a}_2 = \begin{bmatrix} 0 & 1 \\ 6 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} (0 \cdot 4) + (1 \cdot 3) \\ (6 \cdot 4) + (0 \cdot 3) \end{bmatrix} = \begin{bmatrix} 3 \\ 24 \end{bmatrix}$$

$$\alpha_{2,2} = \frac{\mathbf{q}_2^\top \mathbf{k}_2}{\sqrt{d}} = \begin{bmatrix} 4 & 35 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = (4 \cdot 3) + (35 \cdot 24) = 12 + 840 = \frac{852}{\sqrt{2}}$$

$$\mathbf{k}_3 = W_k \mathbf{a}_3 = \begin{bmatrix} 0 & 1 \\ 6 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ 6 \end{bmatrix} = \begin{bmatrix} (0 \cdot -2) + (1 \cdot 6) \\ (6 \cdot -2) + (0 \cdot 6) \end{bmatrix} = \begin{bmatrix} 6 \\ -12 \end{bmatrix}$$

$$\alpha_{2,3} = \frac{\mathbf{q}_2^\top \mathbf{k}_3}{\sqrt{d}} = \begin{bmatrix} 4 & 35 \end{bmatrix} \begin{bmatrix} 6 \\ -12 \end{bmatrix} = (4 \cdot 6) + (35 \cdot -12) = 24 - 420 = \frac{-396}{\sqrt{2}}$$

Solution 2

To write the operation from input $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ to output $\mathbf{b}_2$, we perform the following steps:

$$\begin{aligned}\mathbf{q}_i &= W_q \mathbf{a}_i \\ \mathbf{k}_i &= W_k \mathbf{a}_i \\ \mathbf{v}_i &= W_v \mathbf{a}_i \\ \alpha_{2,i} &= \frac{\mathbf{q}_2^\top \mathbf{k}_i}{\sqrt{d}} \\ \hat{\alpha}_{2,i} &= \frac{\exp(\alpha_{2,i})}{\sum_j \exp(\alpha_{2,j})} \\ \mathbf{Z}_{2,i} &= \hat{\alpha}_{2,i} \cdot \mathbf{v}_i \\ \mathbf{b}_2 &= \sum_i \mathbf{Z}_{2,i} = \hat{\alpha}_{2,1} \cdot \mathbf{v}_1 + \hat{\alpha}_{2,2} \cdot \mathbf{v}_2 + \hat{\alpha}_{2,3} \cdot \mathbf{v}_3\end{aligned}$$

Solution 3

Using multiple heads in a Multi-head Self-Attention mechanism offers several advantages, such as:

- **Enhanced Expressive Power:** Multiple attention heads enable the model to learn diverse relationships between tokens or words within the input. Each head can focus on a unique aspect of the data, allowing the model to simultaneously capture various features. This results in a more comprehensive and nuanced representation of the input, helping the model identify and learn more intricate patterns and connections.
- **Improved Robustness and Generalization:** Each attention head can concentrate on different segments of the input sequence, enabling the model to capture both detailed (local) and broad (global) relationships. This diversity in focus allows the model to perform well across a wide range of tasks and input formats. It also strengthens the model's resistance to distractions such as noise or irrelevant data.
- **Lower Risk of Overfitting:** Multiple attention heads allow the model to learn various attention patterns, which act as a form of regularization. Since each head captures different features of the input, the model becomes less likely to latch onto specific, potentially misleading patterns. This diversity helps the model generalize better, particularly when training data is limited.
- **Parallelization:** Since each attention head functions independently, they can be computed simultaneously. This parallel execution accelerates both

training and inference, improving overall model efficiency—particularly on hardware designed for parallel tasks, such as GPUs and TPUs.

- **Interpretability:** In certain situations, each attention head can be viewed as capturing distinct linguistic or semantic patterns in the input. This helps shed light on the model’s decision-making process and can assist with debugging or analyzing its behavior.

In summary, incorporating multiple heads in multi-head self-attention allows the model to capture more diverse and informative representations, enhances its ability to generalize, helps prevent overfitting, supports efficient parallel computation, and can also improve interpretability.

Problem 2: K-means Clustering

We are given a one-dimensional dataset: $\{5, 7, 16, 18, 24, 26, 34, 38\}$. The objective is to apply K-means clustering with $k = 3$ clusters and address several sub-questions.

1. Clustering Results after Assigning Points to the Nearest Centroid (Initial Centroids: 15, 25, 31)

We are given three initial centroids at $\mu_1 = 15$, $\mu_2 = 25$, and $\mu_3 = 31$. The first step in K-means is to assign each point to the cluster whose centroid is closest.

Step 1: Calculate the distance of each point from the centroids:

Point	Distance to $\mu_1 = 15$	Distance to $\mu_2 = 25$	Distance to $\mu_3 = 31$
5	$ 5 - 15 = 10$	$ 5 - 25 = 20$	$ 5 - 31 = 26$
7	$ 7 - 15 = 8$	$ 7 - 25 = 18$	$ 7 - 31 = 24$
16	$ 16 - 15 = 1$	$ 16 - 25 = 9$	$ 16 - 31 = 15$
18	$ 18 - 15 = 3$	$ 18 - 25 = 7$	$ 18 - 31 = 13$
24	$ 24 - 15 = 9$	$ 24 - 25 = 1$	$ 24 - 31 = 7$
26	$ 26 - 15 = 11$	$ 26 - 25 = 1$	$ 26 - 31 = 5$
34	$ 34 - 15 = 19$	$ 34 - 25 = 9$	$ 34 - 31 = 3$
38	$ 38 - 15 = 23$	$ 38 - 25 = 13$	$ 38 - 31 = 7$

Step 2: Assign each point to the nearest centroid: - Points close to $\mu_1 = 15$: $\{5, 7, 16, 18\}$ - Points close to $\mu_2 = 25$: $\{24, 26\}$ - Points close to $\mu_3 = 31$: $\{34, 38\}$

So, the clusters are: - Cluster A: $\{5, 7, 16, 18\}$ - Cluster B: $\{24, 26\}$ - Cluster C: $\{34, 38\}$

2. Recompute the New Centroids and SSE

Now, recompute the centroids for each cluster:

- New μ_A (for cluster A): Average of $\{5, 7, 16, 18\}$

$$\mu_A = \frac{5 + 7 + 16 + 18}{4} = \frac{46}{4} = 11.5$$

- New μ_B (for cluster B): Average of $\{24, 26\}$

$$\mu_B = \frac{24 + 26}{2} = 25$$

- New μ_C (for cluster C): Average of $\{34, 38\}$

$$\mu_C = \frac{34 + 38}{2} = 36$$

Next, let's compute the SSE:

SSE Calculation:

$$\text{SSE} = \sum_{x_i \in A} (x_i - \mu_A)^2 + \sum_{x_i \in B} (x_i - \mu_B)^2 + \sum_{x_i \in C} (x_i - \mu_C)^2$$

- For A (points $\{5, 7, 16, 18\}$):

$$(5-11.5)^2 + (7-11.5)^2 + (16-11.5)^2 + (18-11.5)^2 = 42.25 + 20.25 + 20.25 + 42.25 = 125$$

- For B (points $\{24, 26\}$):

$$(24 - 25)^2 + (26 - 25)^2 = 1 + 1 = 2$$

- For C (points $\{34, 38\}$):

$$(34 - 36)^2 + (38 - 36)^2 = 4 + 4 = 8$$

$$\text{Total SSE} = 125 + 2 + 8 = 135$$

So, the new centroids are: - $\mu_A = 11.5$ - $\mu_B = 25$ - $\mu_C = 36$

And the overall SSE is 135.

3. Centroids After Convergence and SSE

Now we check if the centroids have changed after the new assignments. We use the same procedure to reassign the points based on the new centroids.

Assigning points based on new centroids: - Points close to $\mu_A = 11.5$: $\{5, 7, 16, 18\}$ - Points close to $\mu_B = 25$: $\{24, 26\}$ - Points close to $\mu_C = 36$: $\{34, 38\}$

It turns out that the assignment hasn't changed. So, the algorithm has converged.

New Centroids: - $\mu_A = 11.5$ - $\mu_B = 25$ - $\mu_C = 36$

SSE after convergence: The SSE is still 135 (as computed previously).

4. Possibility of Empty Clusters

To check whether empty clusters are possible, we look at the following:

- If the initial centroids are chosen such that they are too far apart, it is possible that no points will be assigned to one of the centroids. - For example, if the initial centroids were chosen at positions where no points are near, a cluster could remain empty.

Test Case: If we initialize the centroids at $\mu_1 = 5$, $\mu_2 = 50$, and $\mu_3 = 100$, cluster $\mu_2 = 50$ and $\mu_3 = 100$ would have no points assigned to them because no points exist near these centroids. This would result in empty clusters.

5. Adding 199 Points Between 5 and 7

The new dataset will include the original points along with the 199 equally spaced points between 5 and 7. This will significantly increase the number of points around the centroid $\mu_1 = 15$.

Effect on Clusters: - Points between 5 and 7 will all be closer to $\mu_1 = 15$. - This will likely cause μ_1 to shift closer to the new cluster of points. - $\mu_2 = 25$ and $\mu_3 = 31$ will still remain around their initial positions, but they might get adjusted slightly due to the distribution of the new points.

After running the K-means algorithm again with $k = 3$, the clusters might look like this: - Cluster A: All points near $\mu_1 = 15$, including the 199 new points between 5 and 7. - Cluster B: Points near $\mu_2 = 25$. - Cluster C: Points near $\mu_3 = 31$.

The exact distribution will depend on the final centroid locations after K-means convergence, but the new cluster structure will primarily have a very dense cluster around 5 to 7.

Problem 3: Hierarchical Clustering

Solution 1

We are given a distance matrix between five points labeled p_1, p_2, p_3, p_4, p_5 .

	p_1	p_2	p_3	p_4	p_5
p_1	0	0.3830	0.1474	0.3466	0.1616
p_2	0.3830	0	0.4592	0.1252	0.2227
p_3	0.1474	0.4592	0	0.3840	0.2643
p_4	0.3466	0.1252	0.3840	0	0.2102
p_5	0.1616	0.2227	0.2643	0.2102	0

We are to:

- Perform single-link hierarchical clustering
- Draw the dendrogram, indicating:

- The order in which clusters are merged
- The y-axis showing merge distances

Step-by-step Single Link Hierarchical Clustering

Step 1: Understand Single Link In single link hierarchical clustering, the distance between two clusters is defined as the minimum distance between any pair of points in the two clusters.

This is also known as nearest neighbor clustering.

Step 2: Initialization Start with each point as its own cluster:

$$C = \{\{p_1\}, \{p_2\}, \{p_3\}, \{p_4\}, \{p_5\}\}$$

Step 3: Find and Merge Closest Clusters (Single Link) We will now find the smallest non-zero distance from the matrix and merge the corresponding clusters.

Let's go through the matrix and list all the pairwise distances in increasing order.

From the matrix:

$$\begin{aligned} d(p_3, p_1) = 0.1474, \quad d(p_5, p_1) = 0.1616, \quad d(p_4, p_2) = 0.1252 \quad (\text{smallest}), \\ d(p_5, p_4) = 0.2102, \quad d(p_5, p_2) = 0.2227, \quad d(p_3, p_5) = 0.2643, \dots \end{aligned}$$

Iteration 1: Merge p_2 and $p_4 \rightarrow$ new cluster: $\{p_2, p_4\}$ at distance 0.1252.
Updated clusters:

$$\{\{p_1\}, \{p_2, p_4\}, \{p_3\}, \{p_5\}\}$$

Update the distance matrix between remaining clusters:

We compute the minimum (single link) distance from other clusters to the new cluster $\{p_2, p_4\}$:

$$\text{To } p_1 : \min(d(p_1, p_2), d(p_1, p_4)) = \min(0.3830, 0.3466) = 0.3466$$

$$\text{To } p_3 : \min(d(p_3, p_2), d(p_3, p_4)) = \min(0.4592, 0.3840) = 0.3840$$

$$\text{To } p_5 : \min(d(p_5, p_2), d(p_5, p_4)) = \min(0.2227, 0.2102) = 0.2102$$

New distance matrix:

	p_1	$p_{2,4}$	p_3	p_5
p_1	0	0.3466	0.1474	0.1616
$p_{2,4}$	0.3466	0	0.3840	0.2102
p_3	0.1474	0.3840	0	0.2643
p_5	0.1616	0.2102	0.2643	0

Iteration 2: Smallest distance: p_1 and p_3 at 0.1474.

Merge $\{p_1\}$ and $\{p_3\} \rightarrow$ new cluster: $\{p_1, p_3\}$.

Updated clusters:

$$\{\{p_1, p_3\}, \{p_2, p_4\}, \{p_5\}\}$$

Compute updated distances:

Between $\{p_1, p_3\}$ and $\{p_2, p_4\}$:

$$\min(d(p_1, p_2), d(p_1, p_4), d(p_3, p_2), d(p_3, p_4)) = \min(0.3830, 0.3466, 0.4592, 0.3840) = 0.3466$$

Between $\{p_1, p_3\}$ and p_5 :

$$\min(d(p_1, p_5), d(p_3, p_5)) = \min(0.1616, 0.2643) = 0.1616$$

Between $\{p_2, p_4\}$ and p_5 : already in matrix: 0.2102.

New matrix:

	$p_{1,3}$	$p_{2,4}$	p_5
$p_{1,3}$	0	0.3466	0.1616
$p_{2,4}$	0.3466	0	0.2102
p_5	0.1616	0.2102	0

Iteration 3: Smallest distance: $p_{1,3}$ and p_5 at 0.1616.

Merge $\{p_1, p_3\}$ and $p_5 \rightarrow$ new cluster: $\{p_1, p_3, p_5\}$.

Updated clusters:

$$\{\{p_1, p_3, p_5\}, \{p_2, p_4\}\}$$

Compute updated distances:

Between $\{p_1, p_3, p_5\}$ and $\{p_2, p_4\}$:

$$\min(\min(d(p_1, p_2), d(p_1, p_4), d(p_3, p_2), d(p_3, p_4)), \min(d(p_5, p_2), d(p_5, p_4))) = 0.2102$$

Final matrix:

	$p_{1,3,5}$	$p_{2,4}$
$p_{1,3,5}$	0	0.2102
$p_{2,4}$	0.2102	0

Iteration 4: Merge the final two clusters: $\{p_1, p_3, p_5\}$ and $\{p_2, p_4\}$ at distance 0.2102.

Final cluster: $\{p_1, p_2, p_3, p_4, p_5\}$.

Final Dendrogram Construction

Now let's summarize the hierarchy of merges and their merge distances:

Step	Clusters Merged	Distance
1	$\{p_2, p_4\}$	0.1252
2	$\{p_1, p_3\}$	0.1474
3	$\{p_1, p_3\} + p_5$	0.1616
4	$\{p_1, p_3, p_5\} + \{p_2, p_4\}$	0.2102

This gives us the full dendrogram structure.

Solution 2: Dendrogram by complete link

Initial Distance Matrix

	p_1	p_2	p_3	p_4	p_5
p_1	0	0.3830	0.1474	0.3466	0.1616
p_2	0.3830	0	0.4592	0.1252	0.2227
p_3	0.1474	0.4592	0	0.3840	0.2643
p_4	0.3466	0.1252	0.3840	0	0.2102
p_5	0.1616	0.2227	0.2643	0.2102	0

Step 1: Initialization

Start with each point as its own cluster:

$$C = \{\{p_1\}, \{p_2\}, \{p_3\}, \{p_4\}, \{p_5\}\}$$

Step 2: Find and Merge Closest Clusters

From the matrix:

$$\begin{aligned} d(p_1, p_3) &= 0.1474, & d(p_1, p_5) &= 0.1616, & d(p_2, p_4) &= 0.1252 \text{ (smallest)} \\ d(p_4, p_5) &= 0.2102, & d(p_2, p_5) &= 0.2227, & d(p_3, p_5) &= 0.2643, \dots \end{aligned}$$

Iteration 1: Merge $\{p_2\}$ and $\{p_4\}$

New cluster: $\{p_2, p_4\}$ at distance = 0.1252

Updated clusters: $\{\{p_1\}, \{p_2, p_4\}, \{p_3\}, \{p_5\}\}$

Complete-link distances:

$$\begin{aligned} d(p_1, p_{24}) &= \max(d(p_1, p_2), d(p_1, p_4)) = \max(0.3830, 0.3466) = 0.3830 \\ d(p_3, p_{24}) &= \max(d(p_3, p_2), d(p_3, p_4)) = \max(0.4592, 0.3840) = 0.4592 \\ d(p_5, p_{24}) &= \max(d(p_5, p_2), d(p_5, p_4)) = \max(0.2227, 0.2102) = 0.2227 \end{aligned}$$

	p_1	p_2	p_3	p_5
p_1	0	0.3830	0.1474	0.1616
p_2	0.3830	0	0.4592	0.2227
p_3	0.1474	0.4592	0	0.2643
p_5	0.1616	0.2227	0.2643	0

Iteration 2: Merge $\{p_1\}$ and $\{p_3\}$

New cluster: $\{p_1, p_3\}$ at distance = 0.1474

Updated clusters: $\{\{p_1, p_3\}, \{p_2, p_4\}, \{p_5\}\}$

$$\begin{aligned}
d(p_{13}, p_{24}) &= \max(d(p_1, p_2), d(p_1, p_4), d(p_3, p_2), d(p_3, p_4)) \\
&= \max(0.3830, 0.3466, 0.4592, 0.3840) = 0.4592 \\
d(p_{13}, p_5) &= \max(d(p_1, p_5), d(p_3, p_5)) = \max(0.1616, 0.2643) = 0.2643
\end{aligned}$$

	p_{13}	p_{24}	p_5
p_{13}	0	0.4592	0.2643
p_{24}	0.4592	0	0.2227
p_5	0.2643	0.2227	0

Iteration 3: Merge $\{p_2, p_4\}$ and $\{p_5\}$

New cluster: $\{p_2, p_4, p_5\}$ at distance = 0.2227

Updated clusters: $\{\{p_1, p_3\}, \{p_2, p_4, p_5\}\}$

$$d(p_{13}, p_{245}) = \max(0.3830, 0.3466, 0.4592, 0.3840, 0.2643, 0.2227) = 0.4592$$

	p_{13}	p_{245}
p_{13}	0	0.4592
p_{245}	0.4592	0

Iteration 4: Final Merge

Merge $\{p_1, p_3\}$ and $\{p_2, p_4, p_5\}$ at distance = 0.4592

Final cluster: $\{p_1, p_2, p_3, p_4, p_5\}$

Dendrogram Merge Summary

Step	Merge	Distance
1	$\{p_2\} + \{p_4\}$	0.1252
2	$\{p_1\} + \{p_3\}$	0.1474
3	$\{p_2, p_4\} + \{p_5\}$	0.2227
4	$\{p_1, p_3\} + \{p_2, p_4, p_5\}$	0.4592

Problem 4: Bayes Rule

Given:

- Factory A produces 20% of the phones: $P(A) = 0.20$
- Factory B produces 30% of the phones: $P(B) = 0.30$
- Factory C produces 50% of the phones: $P(C) = 0.50$
- Probability of defective phone from each factory:
 - $P(D|A) = 0.02$ (2% defective from A)
 - $P(D|B) = 0.01$ (1% defective from B)
 - $P(D|C) = 0.0005$ (0.05% defective from C)

Solution 1:

Probability of a phone being defective $P(D)$

We use the Total Probability Theorem:

$$P(D) = P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C)$$

Substituting the given values:

$$P(D) = (0.02)(0.20) + (0.01)(0.30) + (0.0005)(0.50)$$

$$P(D) = 0.004 + 0.003 + 0.00025 = 0.00725$$

Thus, the probability of a defective phone is:

$P(D) = 0.00725 \text{ or } 0.725\%$

Solution 2:

Probability that the defective phone came from factory A $P(A|D)$

We apply Bayes' Theorem:

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)} = \frac{0.02 \cdot 0.20}{0.00725}$$

$$P(A|D) = \frac{0.004}{0.00725} \approx 0.5517$$

Thus, the probability that the defective phone came from factory A is:

$$P(A|D) \approx 0.5517 \text{ or } 55.17\%$$

Solution 3:

Probability that the defective phone came from factory B $P(B|D)$

$$P(B|D) = \frac{P(D|B)P(B)}{P(D)} = \frac{0.01 \cdot 0.30}{0.00725}$$

$$P(B|D) = \frac{0.003}{0.00725} \approx 0.4138$$

Thus, the probability that the defective phone came from factory B is:

$$P(B|D) \approx 0.4138 \text{ or } 41.38\%$$

Solution 4:

Probability that the defective phone came from factory C $P(C|D)$

$$P(C|D) = \frac{P(D|C)P(C)}{P(D)} = \frac{0.0005 \cdot 0.50}{0.00725}$$

$$P(C|D) = \frac{0.00025}{0.00725} \approx 0.0345$$

Thus, the probability that the defective phone came from factory C is:

$$P(C|D) \approx 0.0345 \text{ or } 3.45\%$$

Check: Do the probabilities sum to 1?

$$0.5517 + 0.4138 + 0.0345 = 1.0000 \quad \checkmark$$